

ENGINEERING PHYSICS: COMPREHENSIVE MASTER TEXTBOOK

1. Quantum Mechanics

Wave-Particle Duality

Matter exhibits both wave-like and particle-like properties. De Broglie's hypothesis relates wavelength (λ) to momentum (p).

$$\lambda = h / p = h / (mv)$$

Schrodinger Equation: The fundamental equation describing how the quantum state of a physical system changes with time.

2. Electromagnetism

Maxwell's Equations

These four equations form the foundation of classical electromagnetism, predicting the existence of electromagnetic waves.

Key Concept: Changing magnetic fields induce electric fields (Faraday's Law).

3. Optics and Lasers

Interference and Diffraction

Interference occurs when waves superimpose. Diffraction is the bending of light around obstacles.

Lasers: Operates on the principle of Stimulated Emission of Radiation (population inversion).

4. Long-Answer Problem Bank

Problem 1 (Quantum): Calculate the De Broglie wavelength of an electron (mass = 9.1×10^{-31} kg) moving at 10^6 m/s.

Detailed Solution:

1. Given: $m = 9.1 \times 10^{-31}$ kg, $v = 10^6$ m/s, $h = 6.63 \times 10^{-34}$ J.s.

2. $p = m \cdot v = 9.1 \times 10^{-31} \cdot 10^6 = 9.1 \times 10^{-25}$ kg.m/s.

3. $\lambda = h / p = 6.63 \times 10^{-34} / 9.1 \times 10^{-25} \approx 7.28 \times 10^{-10}$ meters (0.728 nm).

Problem 2 (Optics): In a double-slit experiment, if the wavelength of light is 600nm and slit distance is 0.1mm, find the fringe width on a screen 1m away.

Detailed Solution:

1. Formula: Fringe width (β) = $(\lambda * D) / d$.
2. Given: $\lambda = 600\text{e-}9 \text{ m}$, $D = 1 \text{ m}$, $d = 0.1\text{e-}3 \text{ m}$.
3. $\beta = (600\text{e-}9 * 1) / (0.1\text{e-}3) = 6\text{e-}3 \text{ m} = 6 \text{ mm}$.